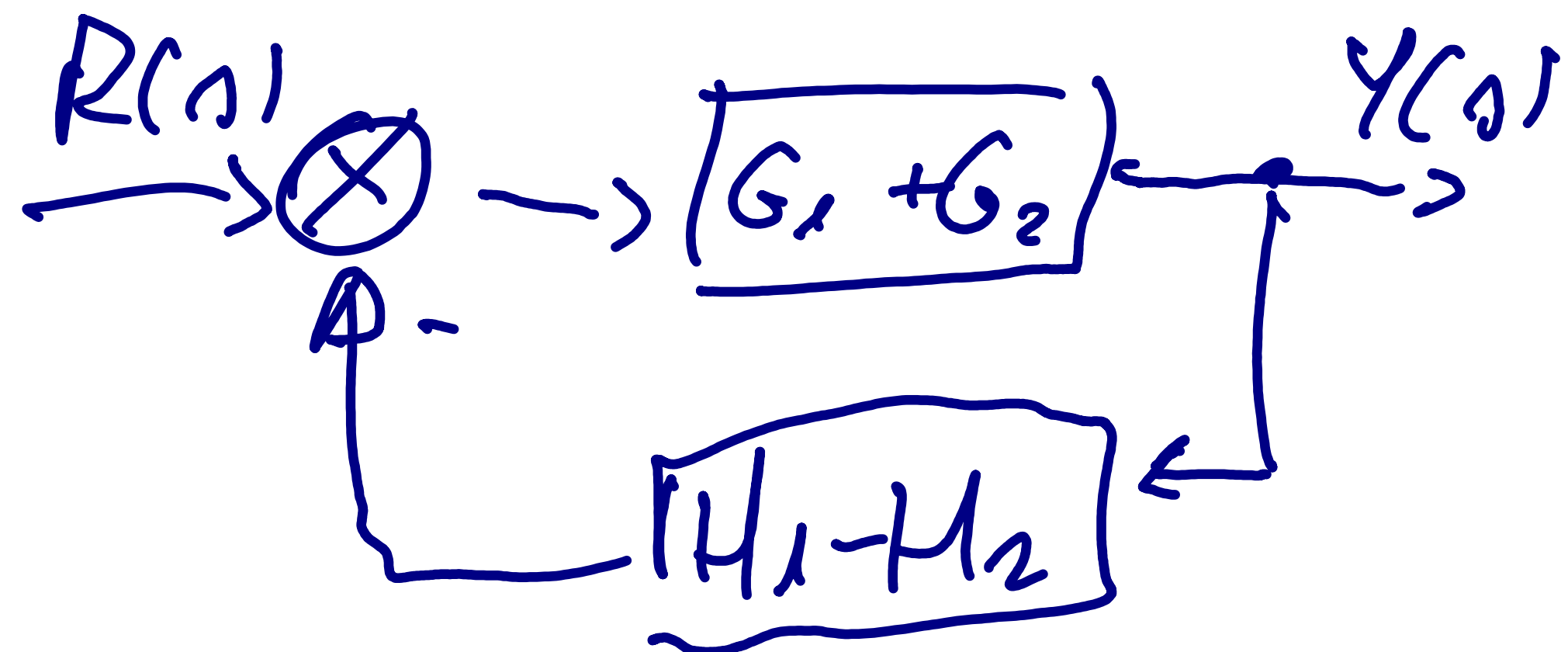
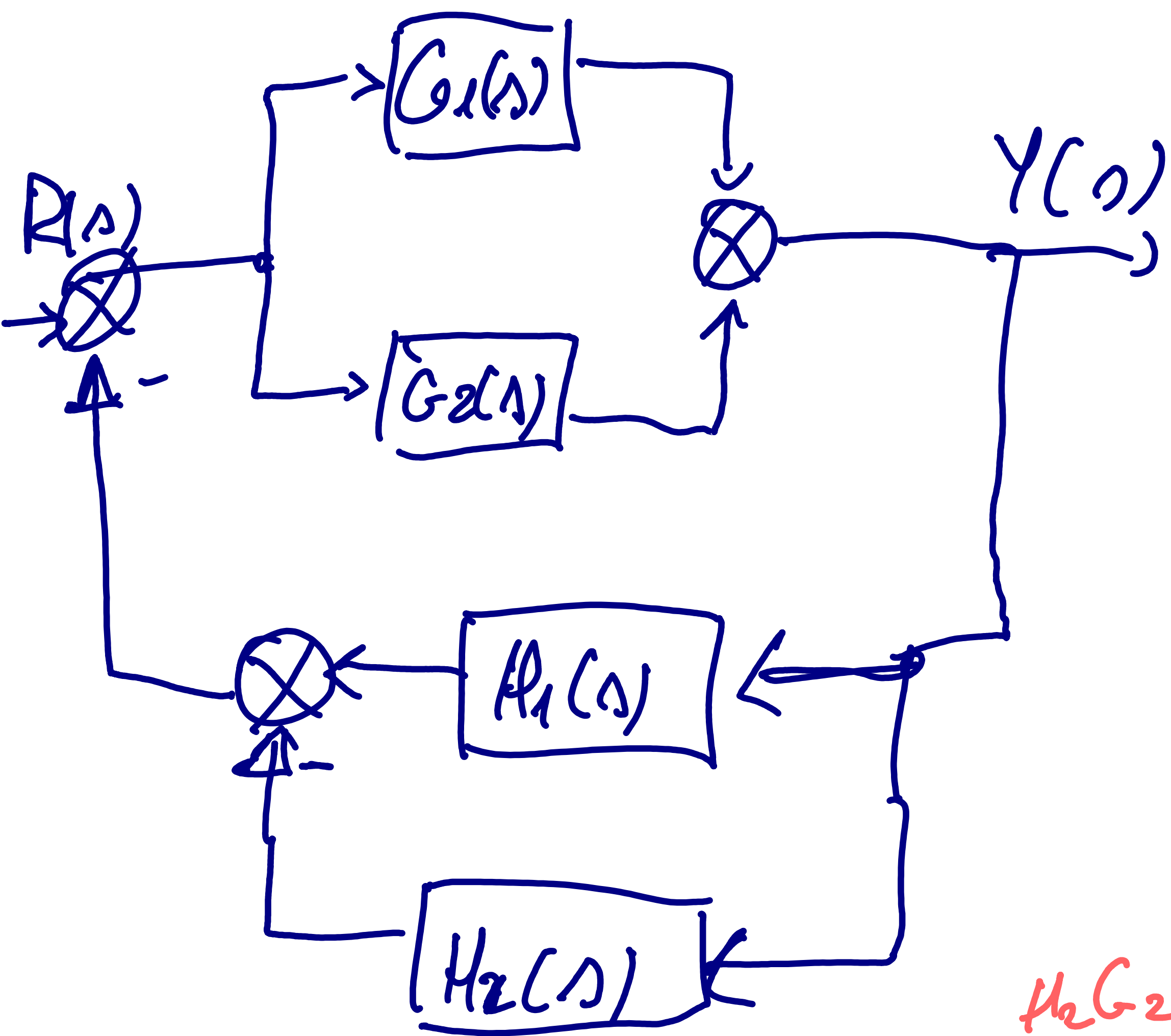
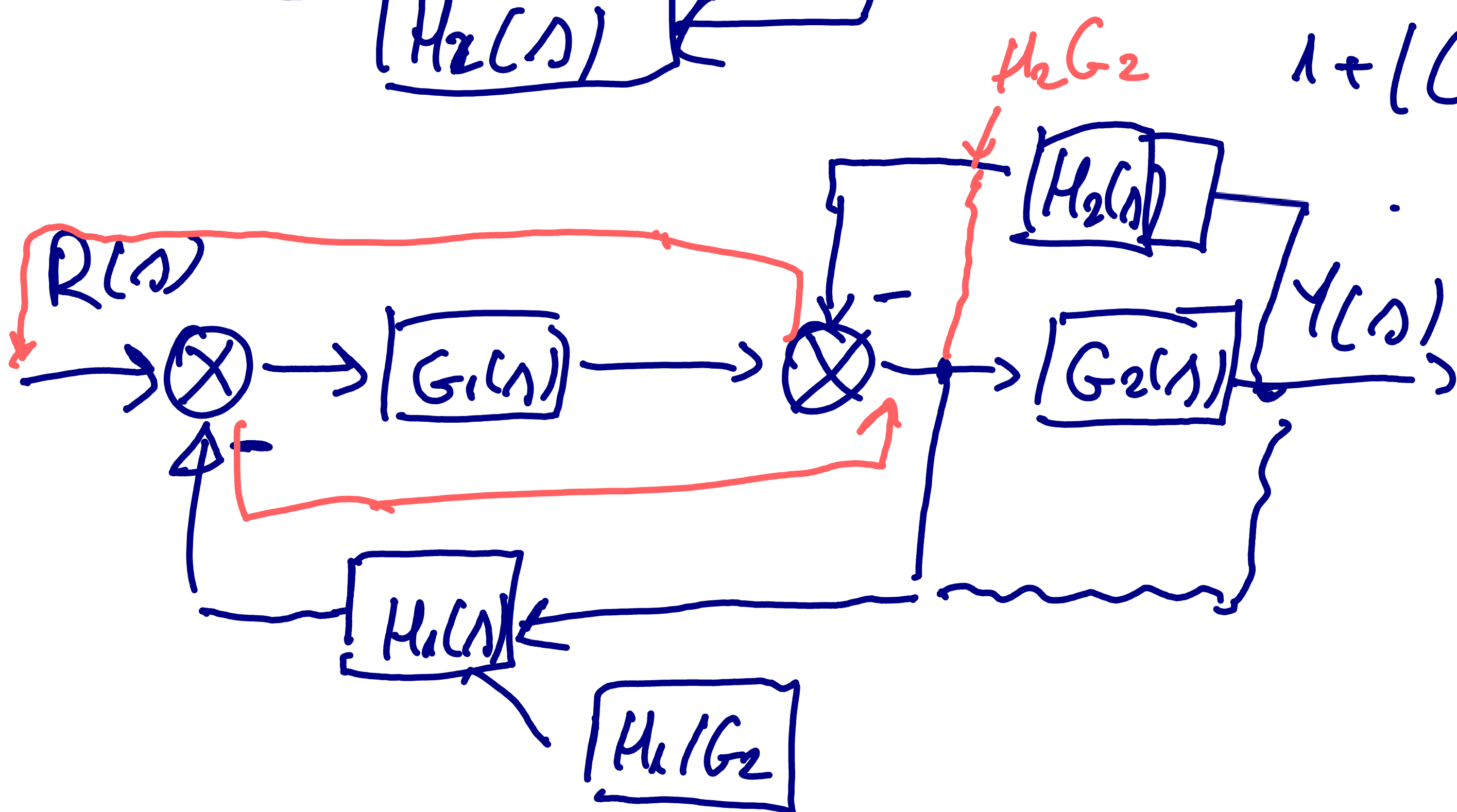


Reducerea schemei bloc



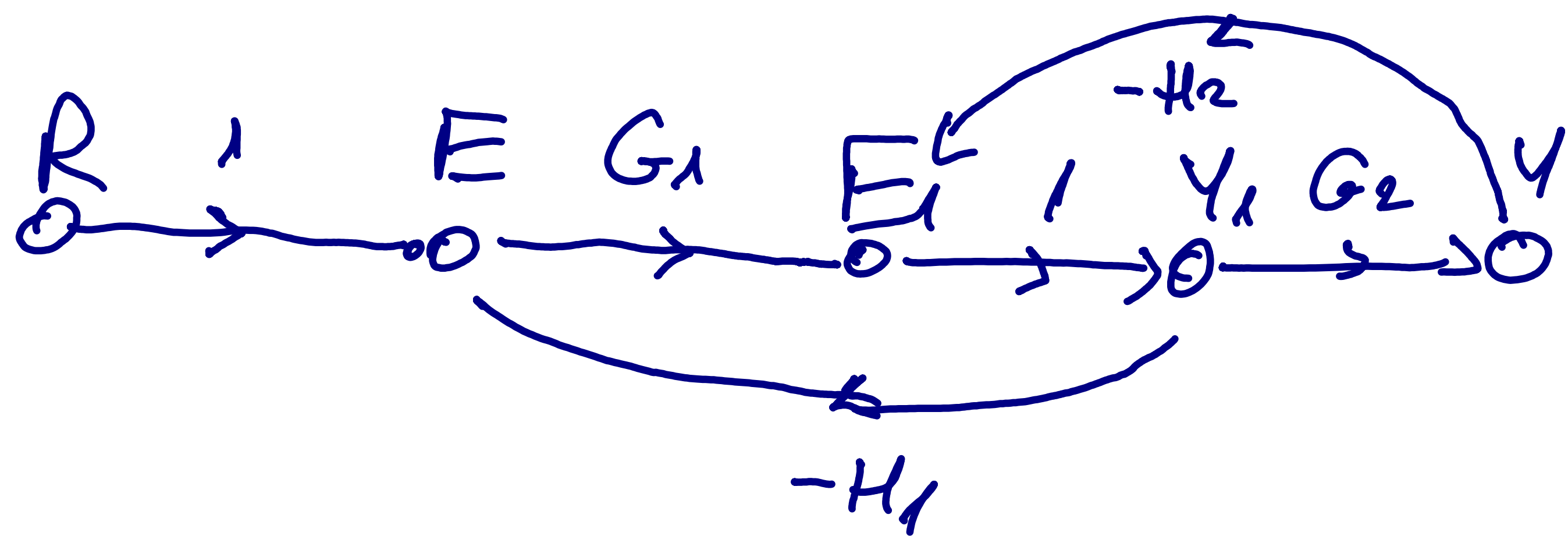
$$G_e(s) = G_o(s) = \frac{G_1(s) + G_2(s)}{1 + (G_1(s) + G_2(s))(H_1(s) + H_2(s))}$$



$$G_{e1}(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$

$$G_{e2}(s) = G_1(s) G_{e1}(s) = \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)}$$

$$G_o(s) = \frac{G_{e2}(s)}{1 + G_{e2}(s) \frac{H_1(s)}{G_2(s)}} = \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)} \frac{1}{1 + \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)} \frac{H_1(s)}{G_2(s)}} \dots$$



$$cd_1 : R \xrightarrow{1} E \xrightarrow{G_1} E_1 \xrightarrow{1} Y_1 \xrightarrow{G_2} Y$$

$$Gcd_1 = G_1 \cdot G_2$$

$$b_1 : E \xrightarrow{G_1} E_1 \xrightarrow{1} Y_1 \xrightarrow{-H_1} E$$

$$Gb_1 = -G_1 H_1$$

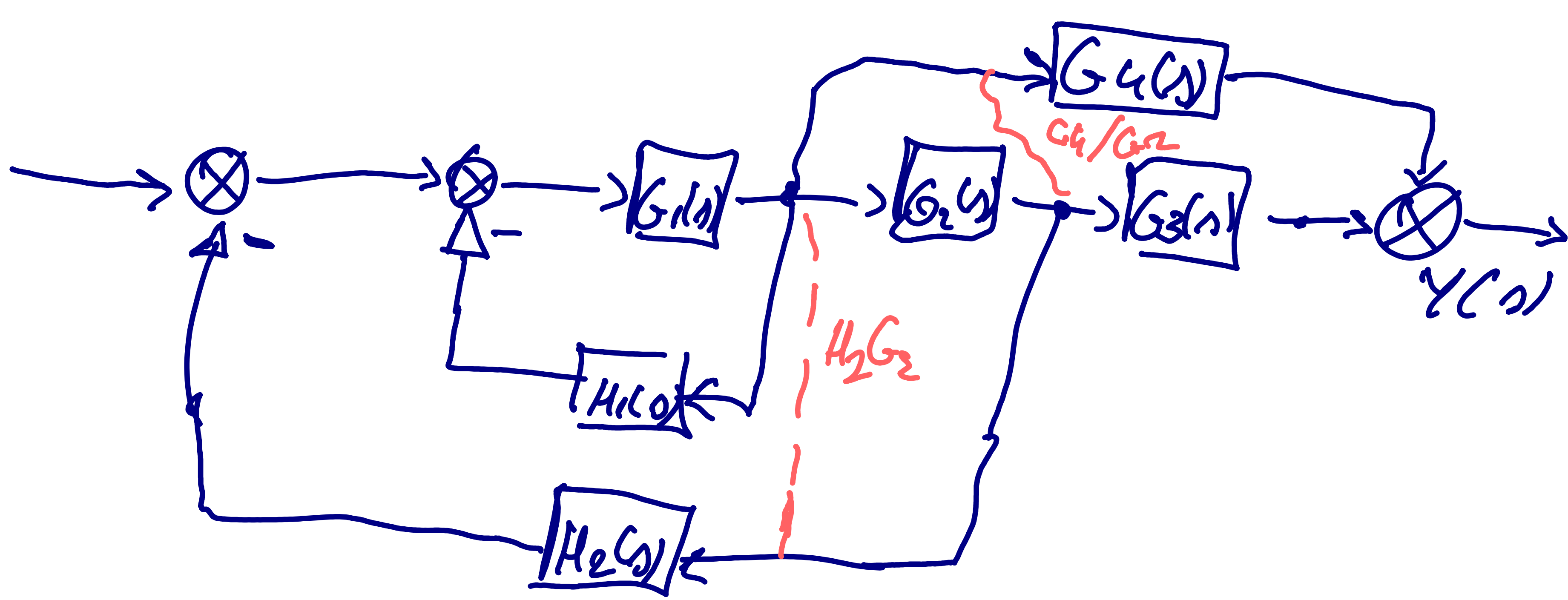
$$b_2 : E_1 \xrightarrow{1} Y_1 \xrightarrow{G_2} Y \xrightarrow{-H_2} E_1$$

$$Gb_2 = -G_2 H_2$$

$$\Delta = 1 - (Gb_1 + Gb_2)$$

$$\Delta_1 = 1$$

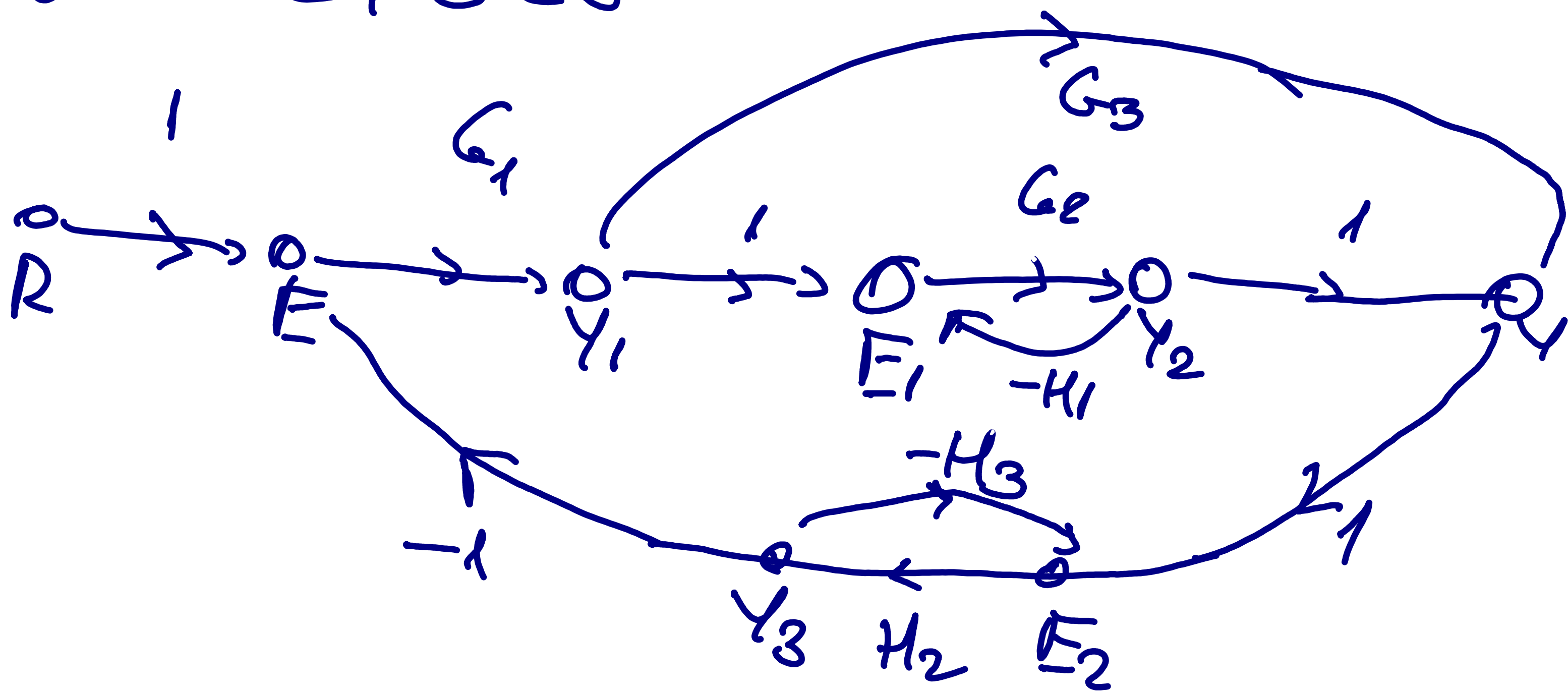
$$G_c = \frac{1}{\Delta} \sum Gcd_i \Delta_i = \frac{1}{\Delta} Gcd_1 \cdot \Delta_1$$



$$G_{e1} = \frac{G_1}{1 + G_1 H_1} \quad G_{e2} = G_{e1} G_2 \quad G_{e3} = \frac{G_{e2}}{1 + G_{e2} H_2}$$

$$G_{e4} = G_{e3} + G_4 / G_2$$

$$G_o = G_{e4} G_3$$



$$cd_1 : R \xrightarrow{1} E \xrightarrow{G_1} Y_1 \xrightarrow{1} E_1 \xrightarrow{G_2} Y_2 \xrightarrow{1} Y \quad G_{cd1} = G_1 G_2$$

$$cd_2 : R \xrightarrow{1} E \xrightarrow{G_1} Y_1 \xrightarrow{G_3} Y \quad G_{cd2} = G_1 G_3$$

$$b_1 : E \xrightarrow{G_1} Y_1 \xrightarrow{1} E_1 \xrightarrow{G_2} Y_2 \xrightarrow{1} Y \xrightarrow{1} E_2 \xrightarrow{H_2} Y_3 \xrightarrow{-1} E$$

$$G_{b1} = -G_1 G_2 H_2$$

$$b_2 : E_2 \xrightarrow{H_2} Y_3 \xrightarrow{-H_3} E_2 \quad G_{b2} = -H_2 H_3$$

$$b_3 : E_1 \xrightarrow{G_2} Y_2 \xrightarrow{-H_1} E_1 \quad G_{b3} = -H_1 G_2$$

$$b_4 : E \xrightarrow{G_1} Y_1 \xrightarrow{G_3} Y \xrightarrow{1} E_2 \xrightarrow{H_2} Y_3 \xrightarrow{-1} E$$

$$Gb_4 = -G_1 G_3 H_2$$

$$\Delta = 1 - (Gb_1 + Gb_2 + Gb_3 + Gb_4) + (Gb_2 Gb_3 + Gb_3 Gb_4)$$

$$\Delta_1 = 1 - Gb_2$$

$$\Delta_2 = 1 - Gb_2 - Gb_3 + Gb_2 Gb_3$$